

NAME Steven P. Smith TITLE Design I - CFI & Ca DWG. NO. 3/23/25

SMITH CHART FORM ZY-01-N COLOR BY J. COLVIN, UNIVERSITY OF FLORIDA, 1997

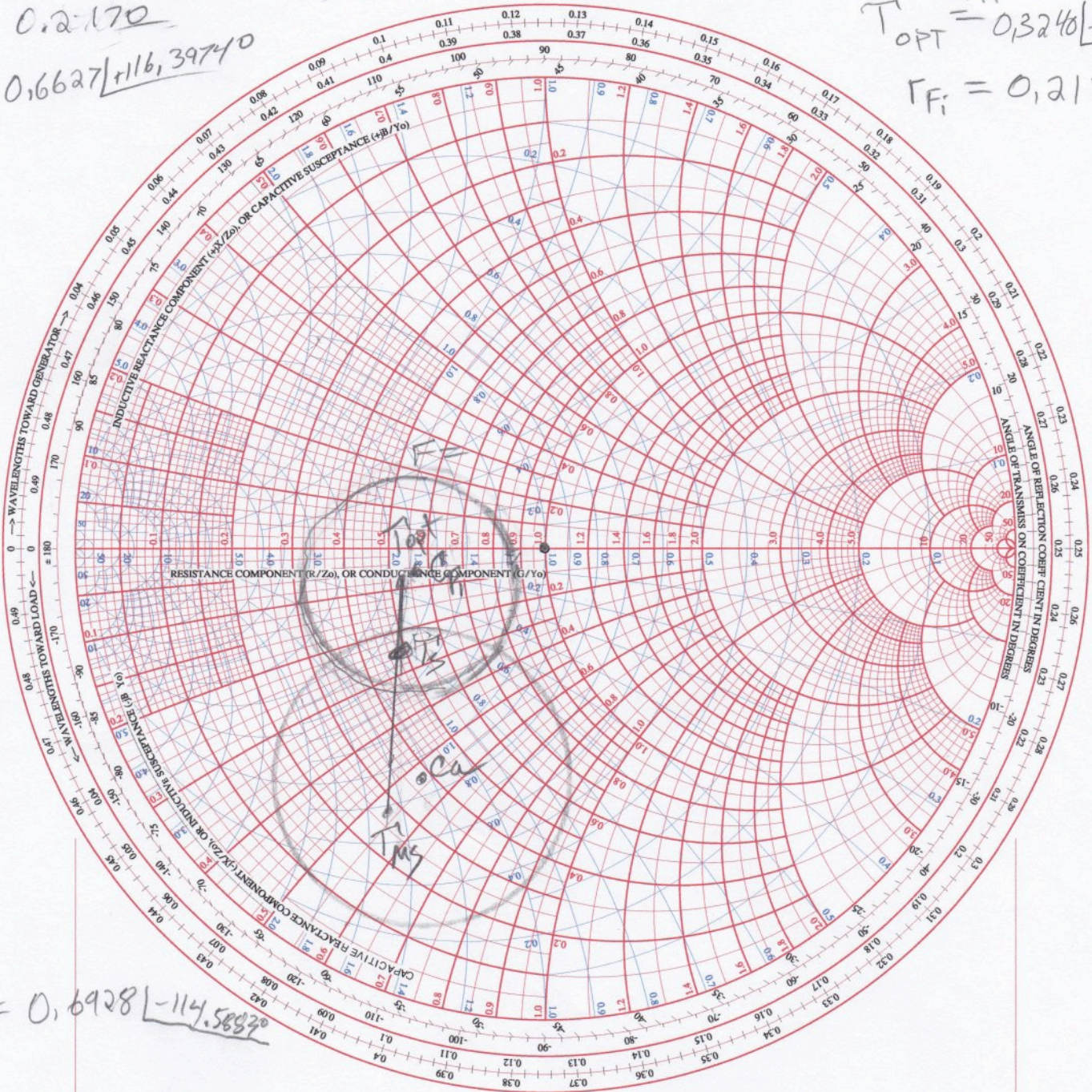
Design Project #2

1

$C_a = (0.6928 \angle -114.5883^\circ)$   
 $r_a = 0.2170$   
 $T_{MS} = 0.6627 \angle 116.3974^\circ$

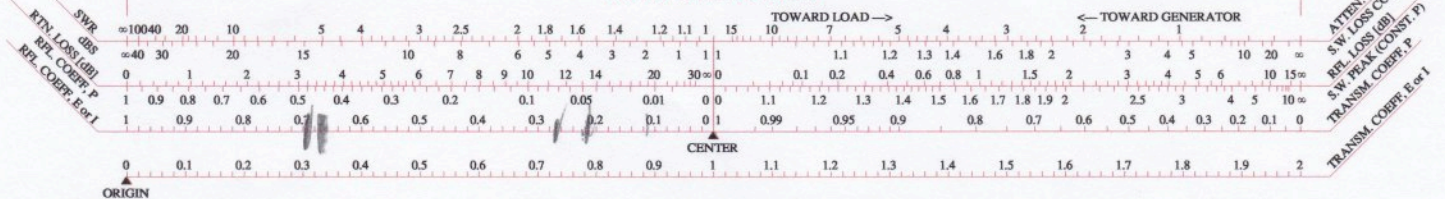
NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

$C_{Fi} = (0.3071 \angle -165.15^\circ)$   
 $T_{OPT} = 0.3240 \angle -165.1508^\circ$   
 $\Gamma_{Fi} = 0.2170$



$T_z = 0.6928 \angle -114.5883^\circ$

RADIALLY SCALED PARAMETERS





# EE 456 - Design Project 2

|                               |                                                 |                        |
|-------------------------------|-------------------------------------------------|------------------------|
| NAME<br><i>Stavros Pappas</i> | TITLE<br><i>Design 2</i>                        | DWG. NO.               |
| SMITH CHART FORM ZY-01-N      | COLOR BY J. COLVIN, UNIVERSITY OF FLORIDA, 1997 | DATE<br><i>3/25/25</i> |

L2

$$\Gamma_{Fi} = 0.2170$$

$$\Gamma_{Fi} = 0.15431 \angle -165.1500^\circ$$

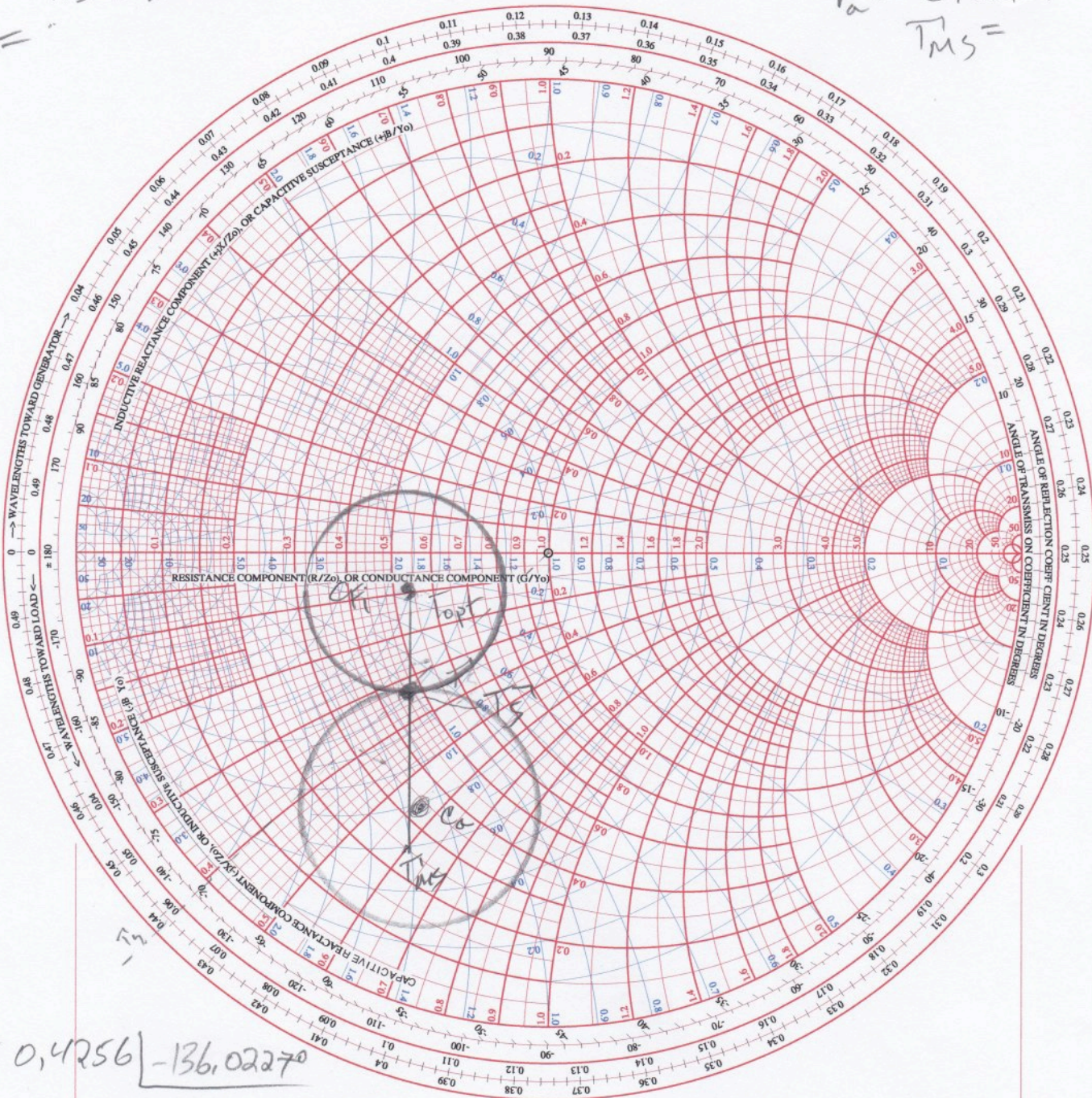
NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

$$C_a = 0.5905 \angle -114.50^\circ$$

$$\Gamma_a = 0.3251$$

$$\Gamma_{MS} =$$

$$\Gamma_{OPT} =$$

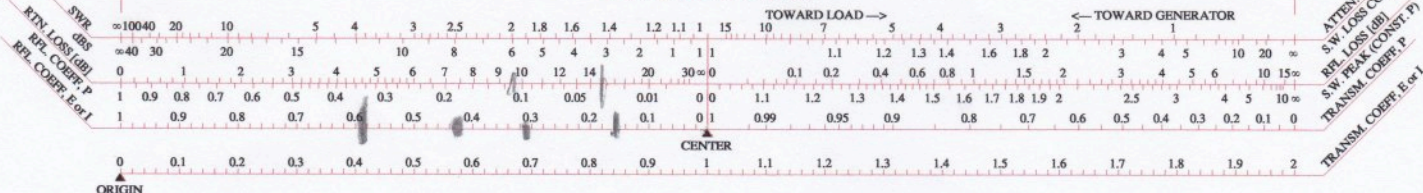


$$\Gamma_s = 0.4256 \angle -136.0227^\circ$$

RADIALLY SCALED PARAMETERS

TOWARD LOAD →

← TOWARD GENERATOR





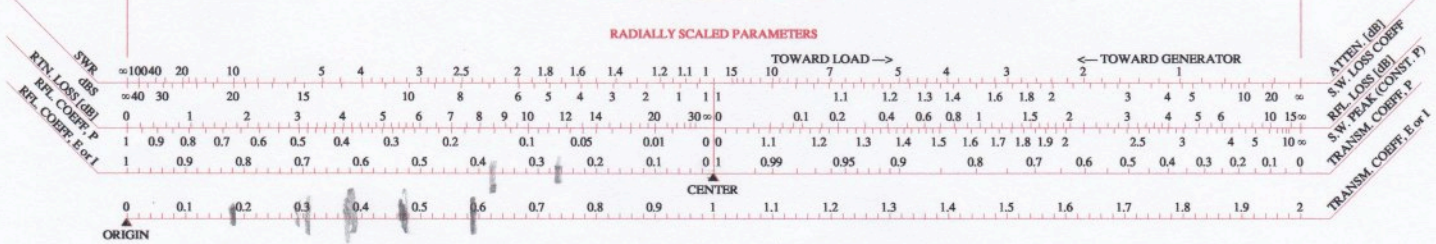
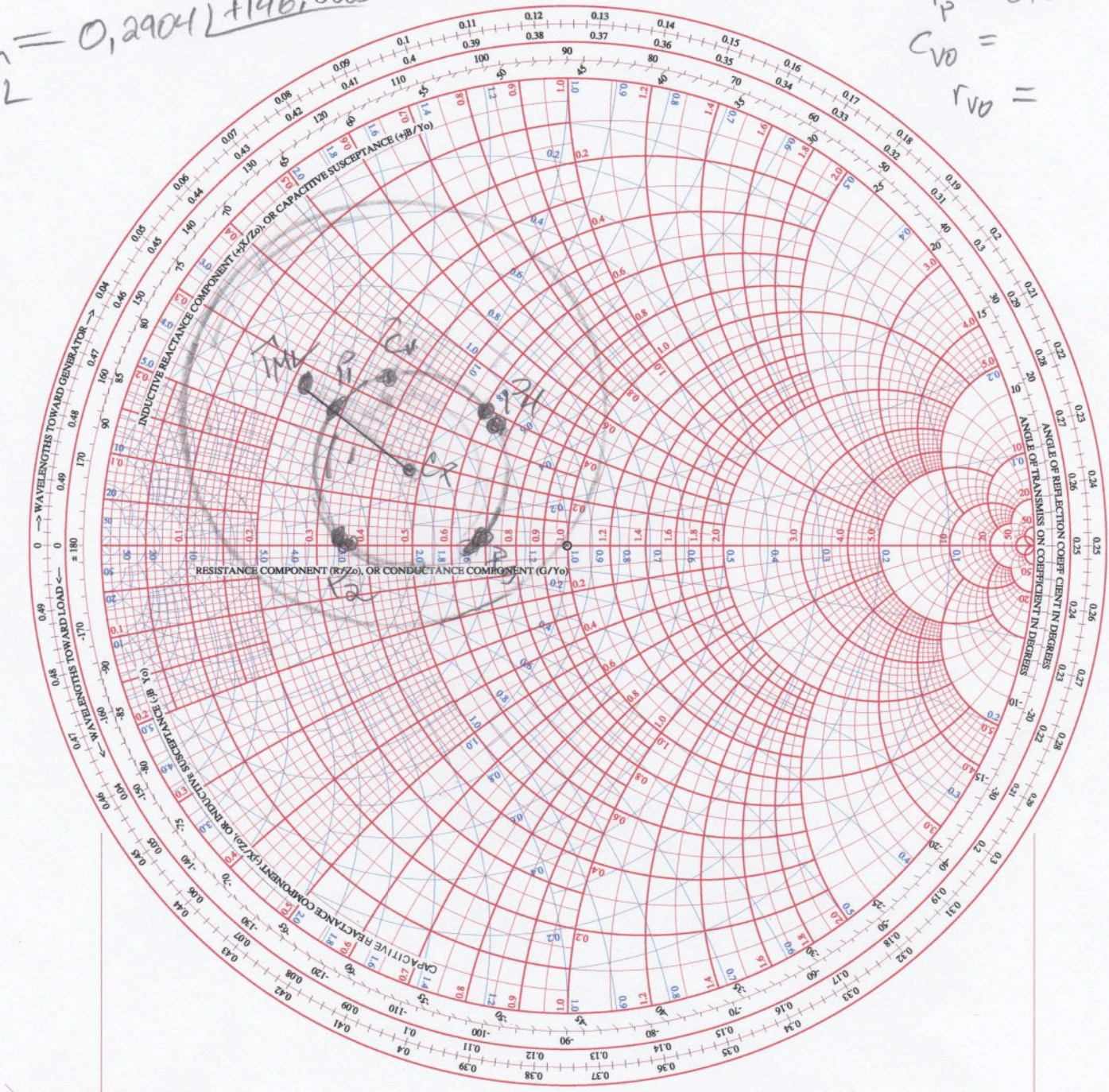
|                            |                                                 |          |
|----------------------------|-------------------------------------------------|----------|
| NAME<br><i>Steven Pung</i> | TITLE<br><i>Design Two</i>                      | DWG. NO. |
| SMITH CHART FORM ZY-01-N   | COLOR BY J. COLVIN, UNIVERSITY OF FLORIDA, 1997 | DATE     |

L3

# NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

$T_{ML} =$   
 $P_{3T_2} = 0.2904 \angle 146.0865^\circ$

$C_p = 0.4646 \angle 152.4988^\circ$   
 $r_p = 0.3251$   
 $C_{v0} =$   
 $r_{v0} =$





NAME  
*Steven Pungk*  
SMITH CHART FORM ZY-01-N

TITLE  
*EE456 Design 2*  
*IMN Part 1*  
COLOR BY J. COLVIN, UNIVERSITY OF FLORIDA, 1997

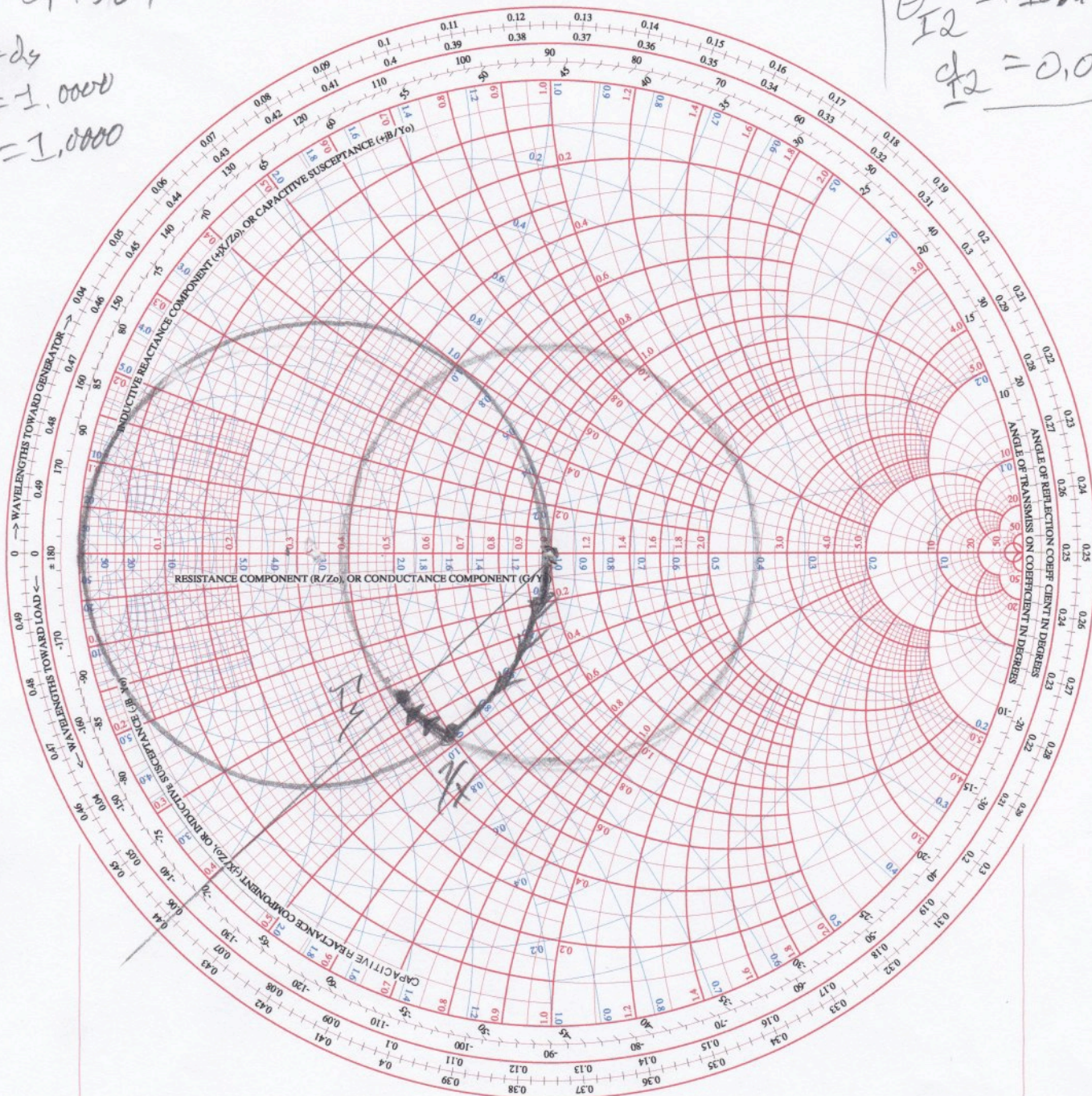
DWG. NO.  
DATE



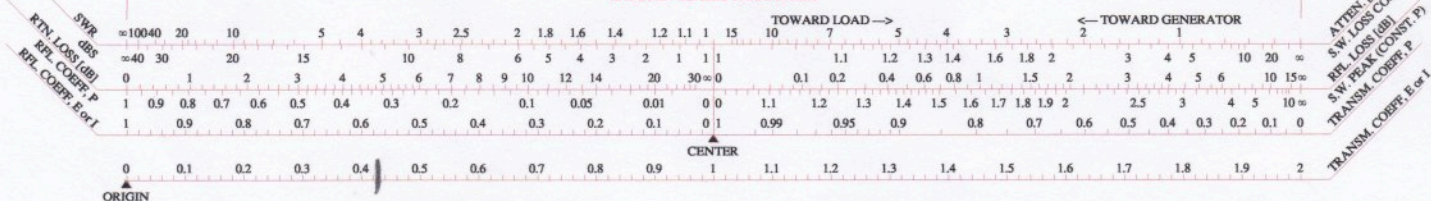
$T_s = 0.4256 \angle -136.02270$   
 $d_s = 0.4384 \lambda \text{ WTG}$   
 $d_{\text{stop}} = d_s$   
 $Z_T = 1.0000$   
 $Y_T = 1.0000$

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

$\theta_{I2} = 110.4170^\circ$   
 $d_{I2} = 0.0318 \lambda$



RADIALLY SCALED PARAMETERS





NAME *Shenn Park*

TITLE *IMN Part 2*

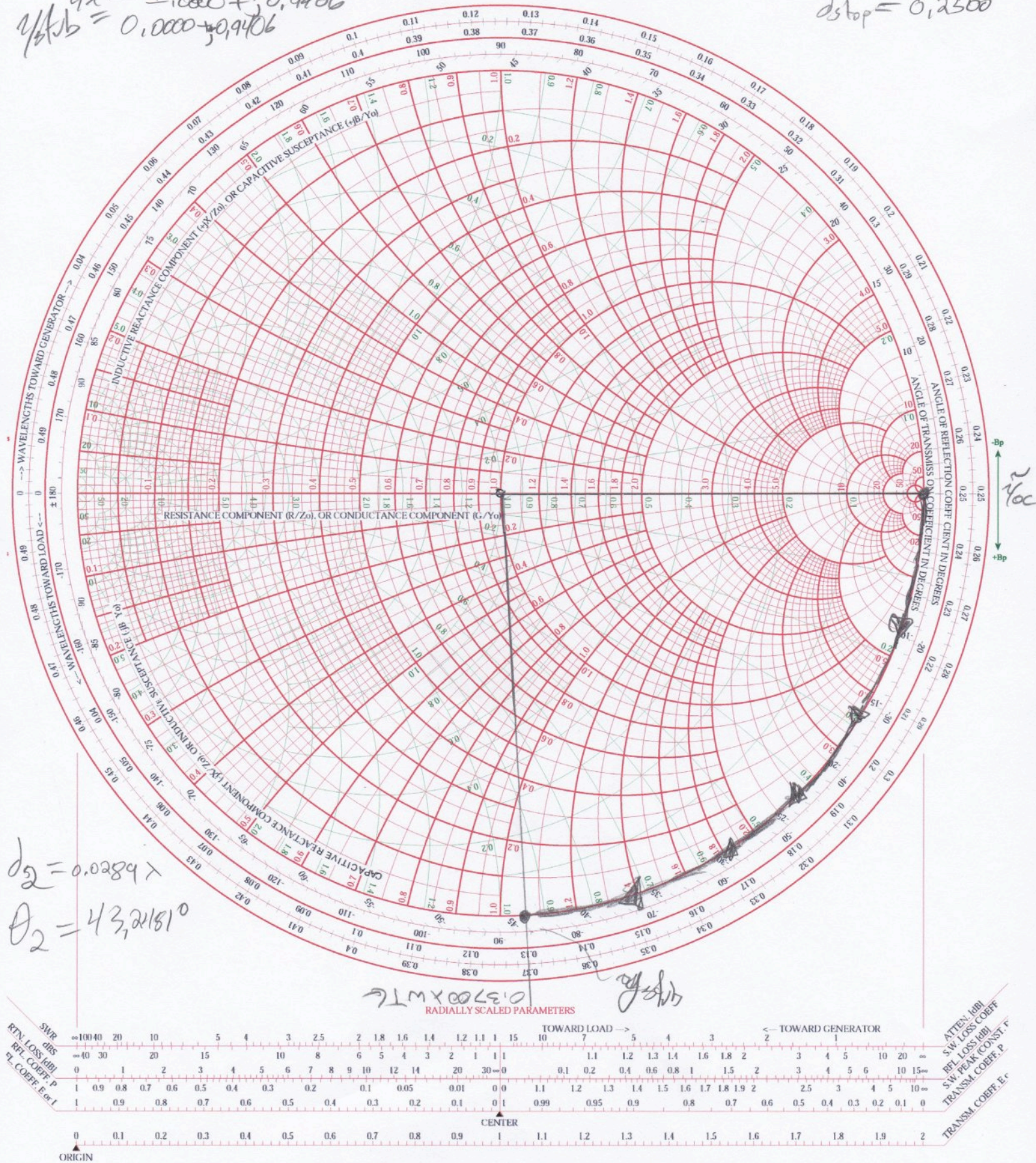
DWG. NO. *303/25*  
DATE

*15*

$Z_x = 0.5306 - j0.4991$   
 $Y_x = 1.0000 + j0.9406$   
 $Y_{fb} = 0.0000 + j0.9406$

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

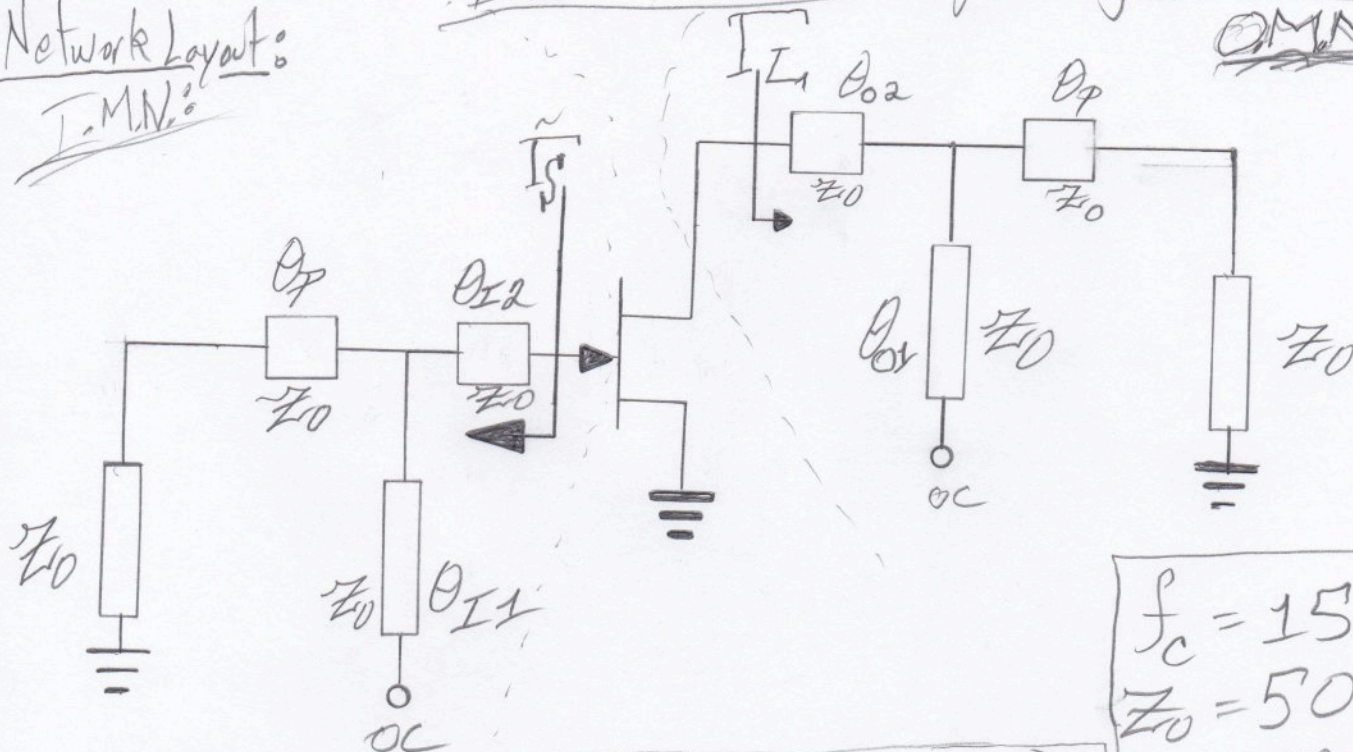
$d_{start} = d_{ac} = 0.2500$   
 $d_{stop} = 0.2500$





Network Layout:  
I.M.N.:

OMN.:



$$\Gamma_L = (0.2904 \angle +146.0865^\circ)$$

$$\Gamma_S = (0.7256 \angle -136.0227^\circ)$$

$$f_c = 15 \text{ GHz}$$

$$Z_0 = 50 \Omega$$

$$\theta_p = 90.0000^\circ$$

$$d_{I1} = 0.0318 \lambda$$

$$\theta_{I1} = 10.4170^\circ$$

$$d_{I2} = 0.0289 \lambda$$

$$\theta_{I2} = 43.2481^\circ$$

$$d_{o1} = 0.0868 \lambda$$

$$\theta_{o1} = 31.2559^\circ$$

$$d_{o2} = 0.1487 \lambda$$

$$\theta_{o2} = 53.5158^\circ$$



|                           |                                                                     |                        |
|---------------------------|---------------------------------------------------------------------|------------------------|
| NAME<br><i>Ham Pangek</i> | TITLE<br><i>EE 456 Design Project 2</i><br><i>OMN Design Part 1</i> | DWG. NO.               |
|                           |                                                                     | DATE<br><i>3/23/25</i> |

17

$\Gamma_L = 0.2904 \angle 146.0865^\circ$

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

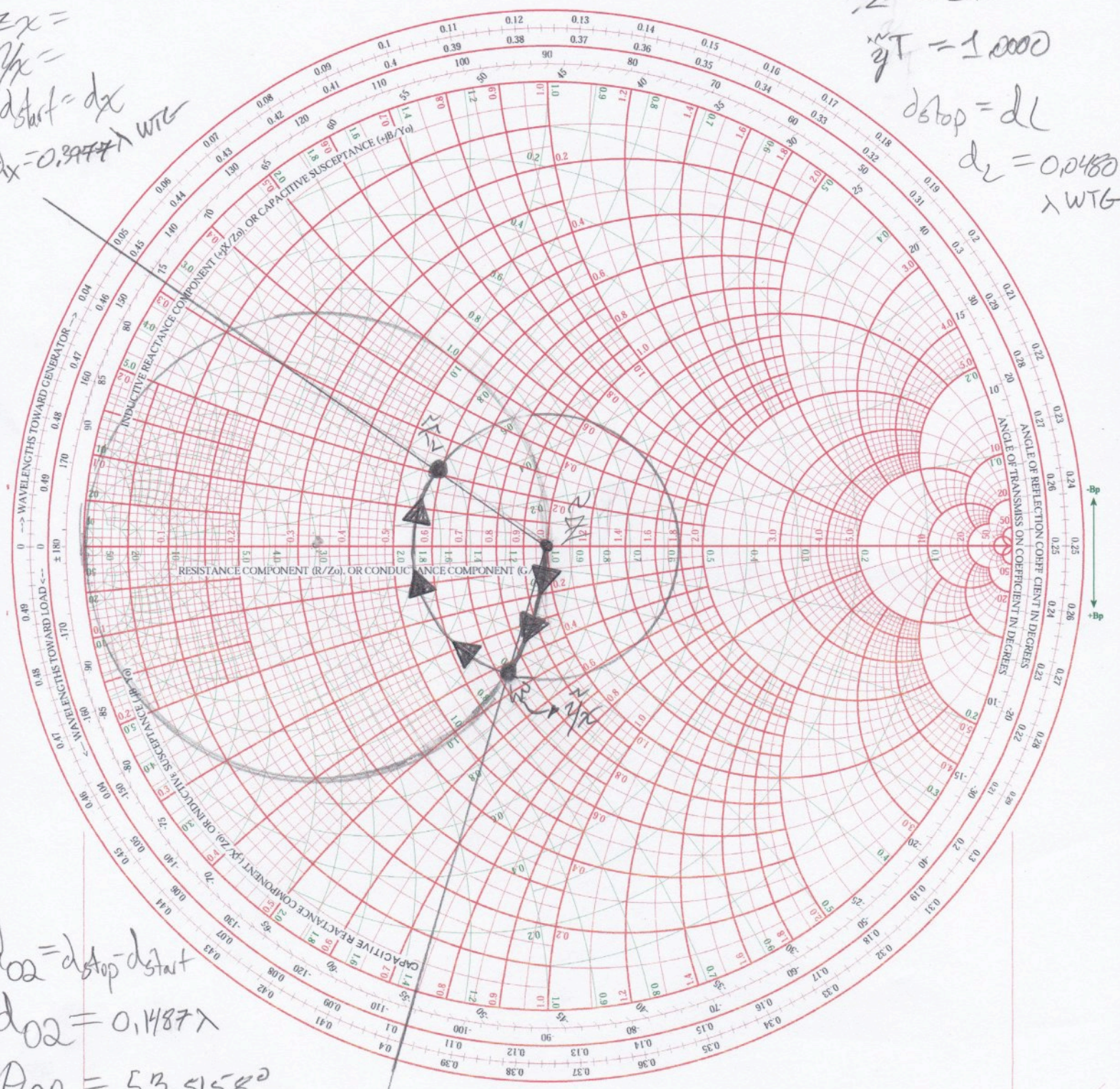
$\frac{V}{Z} = 1.0000$

$\frac{I}{Y} = 1.0000$

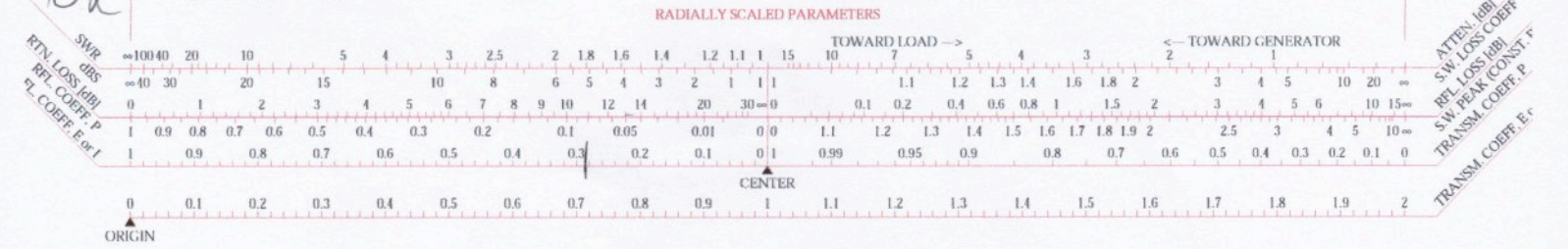
$d_{stop} = d_L$

$d_L = 0.1480 \lambda \text{ WTG}$

$z_x =$   
 $y_x =$   
 $d_{start} = d_x$   
 $d_x = 0.3977 \lambda \text{ WTG}$



$d_{02} = d_{stop} - d_{start}$   
 $d_{02} = 0.1487 \lambda$   
 $\theta_{02} = 53.5158^\circ$





|                              |                                   |                        |
|------------------------------|-----------------------------------|------------------------|
| NAME<br><i>Steven Pangok</i> | TITLE<br><i>OMN Design Part 2</i> | DWG. NO.               |
|                              |                                   | DATE<br><i>3/23/25</i> |

18

$d_{o1} = (0.3368 - 0.25)$   
 $d_{o1} = 0.0868$   
 $\theta_{o1} = 31.2559^\circ$

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

$y_{oc} = 0$   
 $z_{oc} = \infty$   
 $d_{oc} = 0.2500$   
 $d_{start} = d_{oc}$   
 $y_{stub} = 90.6070$   
 $z_{stub} = -j1.6476$   
 $d_{stub} = 0.368$   
 $d_{stop} = d_{stub}$

